



Sir Syed
C@SE Institute of Technology, Islamabad
Department of Computer Science

Assignment 4

Due Date: May 26,2026

Course Title:	Web Engineering	Course Code	CSC3008	Credit Hours:	3(2,1)
Instructor:	Mr. Muhammad Hassan Shah	Program Name:	BS (CYS/CS)		
Instructions for Students: • Do not copy code directly — understand and implement • Keep your code clean and well-structured • Focus on quality over quantity • Late submissions may result in mark deductions					

API & Axios

[CLO2, CLO3]

Q1. Fetch and Display Data Make a GET request using Axios to <https://jsonplaceholder.typicode.com/users> and display the list of user names and emails in an HTML page.

Q2. Post a New Resource Using Axios, send a POST request to <https://jsonplaceholder.typicode.com/posts> with a title, body, and userId. Log the created resource response to the console.

Q3. Handle 404 Errors Make a GET request to an invalid endpoint and use Axios error handling to display a user-friendly error message on the page instead of crashing.

Q4. CRUD Operations Build a simple UI that performs all four CRUD operations (GET, POST, PUT, DELETE) on <https://jsonplaceholder.typicode.com/posts>. Each operation should be triggered by a button click.

Q5. Axios Interceptors Set up Axios request and response interceptors that:

- a. Attach a fake Authorization: Bearer <token> header to every request
- b. Log the response status code for every response

Q6. Search with Query Parameters Build a user search feature that sends a GET request with query parameters (e.g., ?q=john) to a public API like the GitHub Users API and displays results dynamically as the user types (with debouncing).

Q7. Axios Instance with Base URL Create a reusable Axios instance configured with a base URL, default headers, and a timeout of 5 seconds. Use it to make at least three different API calls in a single app.

Q8. Concurrent API Requests Use `axios.all()` (or `Promise.all`) to fetch data from three different endpoints simultaneously — posts, users, and todos from JSONPlaceholder — and display them on the same page only after all three have loaded.

Q9. File Upload with Progress Bar Build a form that uploads an image file to a mock API using Axios. Use the `onUploadProgress` config option to display a real-time progress bar showing the upload percentage.

Q10. Retry Logic on Failure Implement a custom Axios wrapper that automatically retries a failed request up to 3 times with a 2-second delay between attempts. Demonstrate it by calling an unreliable or rate-limited endpoint.